

PVC cutting / choose one gate size

Workshop summary. Print the following page for your chosen opening and paper width. Diagrams are not full-size templates.

Clear opening	Paper width	8 long pieces	16 short pieces	Ratchet sticks	Saw plan*
1000 mm	24 in	1080.0 mm	389.6 mm	5	6
1000 mm	18 in	1080.0 mm	237.2 mm	5	5
1000 mm	12 in	1080.0 mm	84.8 mm	4	4
800 mm	24 in	880.0 mm	389.6 mm	5	5
800 mm	18 in	880.0 mm	237.2 mm	4	4
800 mm	12 in	880.0 mm	84.8 mm	3	3

Do not subtract socket insertion from these numbers. Each end inserts 30 mm into its fitting. The cut lengths already include that engagement. They are neither exposed spans nor node-center distances.

For your ratcheting cutter: layouts allow 10 mm per 3048 mm stick for end cleanup, with no saw kerf. Square the starting end, measure each next piece from the new cut end, cut on the waste side of the mark, then check the finished length. Avoid cumulative marks along a full stick.

Before batch cutting: make one short and one long piece, remove burrs, mark 30 mm insertion at each end and dry-fit with the accepted fittings. Use the first accurate piece as a comparison master for repeats. For 12-inch paper the short piece is only 84.8 mm: check that trial corner carefully.

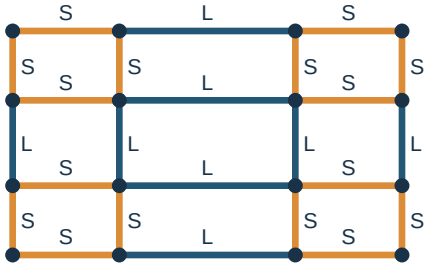
*The existing saw layouts reserve 10 mm per stick plus 3 mm per cut. With the ratchet, the 1000 mm / 24-inch version needs **five sticks instead of six**. The other stock counts are unchanged. Confirm actual stick length; trim losses beyond the allowance require another layout or spare stock.

These quantities cover the flat gate grid only; feet, ballast structures and other ground supports are additional. All six use the same printed fittings. Label long pieces L1-L8 and short pieces S1-S16.

1000 mm opening / 24-inch paper

Outside paper square: 2219.2 mm. Buy 5 sticks of 10-ft nominal 1-inch PVC for the gate grid.

Mark	Cut quantity	Finished length (mm)	Decimal inches
L	8	1080.0	42.5197
S	16	389.6	15.3386



Front grid / schematic, not to scale

Ratcheting cutter / stick-by-stick allocation

Blue = long (L). Orange = short (S). Gray = spare after 10 mm cleanup reserve. Numbers are quantities.

Stick 1	2 L	2 S	2 L + 2 S	98.8 mm spare
Stick 2	2 L	2 S	2 L + 2 S	98.8 mm spare
Stick 3	2 L	2 S	2 L + 2 S	98.8 mm spare
Stick 4	1 L	5 S	1 L + 5 S	10.0 mm spare
Stick 5	1 L	5 S	1 L + 5 S	10.0 mm spare

Finished-piece check:

Long: 8

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Short: 16

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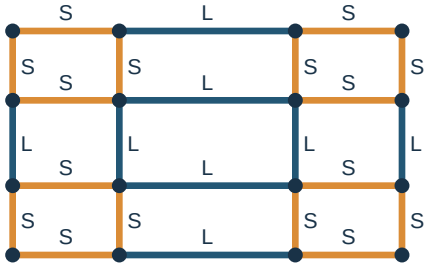
Assembly check: 30 mm insertion at each end. Short-section exposed span = 329.6 mm; long-section exposed span = 1020.0 mm. Finished gate opening = 1000 mm square.

For a hand saw, use the conservative stock-layout CSV linked in the guide; never add saw kerf to a finished part dimension.

1000 mm opening / 18-inch paper

Outside paper square: 1914.4 mm. Buy 5 sticks of 10-ft nominal 1-inch PVC for the gate grid.

Mark	Cut quantity	Finished length (mm)	Decimal inches
L	8	1080.0	42.5197
S	16	237.2	9.3386



Front grid / schematic, not to scale

Ratcheting cutter / stick-by-stick allocation

Blue = long (L). Orange = short (S). Gray = spare after 10 mm cleanup reserve. Numbers are quantities.

Stick 1	2 L	3 S	2 L + 3 S	166.4 mm spare
Stick 2	2 L	3 S	2 L + 3 S	166.4 mm spare
Stick 3	2 L	3 S	2 L + 3 S	166.4 mm spare
Stick 4	2 L	3 S	2 L + 3 S	166.4 mm spare
Stick 5	4 S		0 L + 4 S	2089.2 mm spare

Finished-piece check:

Long: 8

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Short: 16

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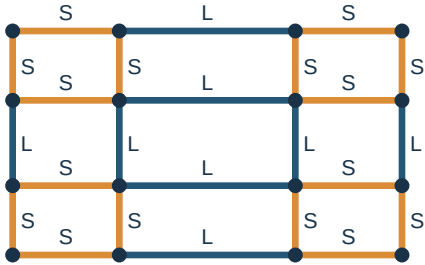
Assembly check: 30 mm insertion at each end. Short-section exposed span = 177.2 mm; long-section exposed span = 1020.0 mm. Finished gate opening = 1000 mm square.

For a hand saw, use the conservative stock-layout CSV linked in the guide; never add saw kerf to a finished part dimension.

1000 mm opening / 12-inch paper

Outside paper square: 1609.6 mm. Buy 4 sticks of 10-ft nominal 1-inch PVC for the gate grid.

Mark	Cut quantity	Finished length (mm)	Decimal inches
L	8	1080.0	42.5197
S	16	84.8	3.3386



Front grid / schematic, not to scale

Ratcheting cutter / stick-by-stick allocation

Blue = long (L). Orange = short (S). Gray = spare after 10 mm cleanup reserve. Numbers are quantities.

Stick 1	2 L	4 S	2 L + 4 S	538.8 mm spare
Stick 2	2 L	4 S	2 L + 4 S	538.8 mm spare
Stick 3	2 L	4 S	2 L + 4 S	538.8 mm spare
Stick 4	2 L	4 S	2 L + 4 S	538.8 mm spare

Finished-piece check:

Long: 8

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Short: 16

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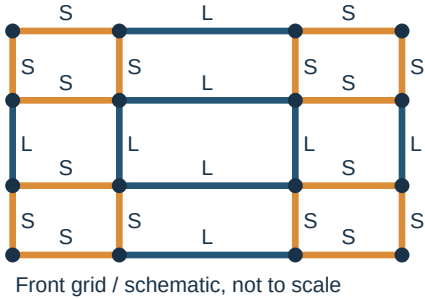
Assembly check: 30 mm insertion at each end. Short-section exposed span = 24.8 mm; long-section exposed span = 1020.0 mm. Finished gate opening = 1000 mm square.

For a hand saw, use the conservative stock-layout CSV linked in the guide; never add saw kerf to a finished part dimension.

800 mm opening / 24-inch paper

Outside paper square: 2019.2 mm. Buy 5 sticks of 10-ft nominal 1-inch PVC for the gate grid.

Mark	Cut quantity	Finished length (mm)	Decimal inches
L	8	880.0	34.6457
S	16	389.6	15.3386



Ratcheting cutter / stick-by-stick allocation

Blue = long (L). Orange = short (S). Gray = spare after 10 mm cleanup reserve. Numbers are quantities.

Stick 1	3 L 1 S	3 L + 1 S	8.4 mm spare
Stick 2	3 L 1 S	3 L + 1 S	8.4 mm spare
Stick 3	1 L 5 S	1 L + 5 S	210.0 mm spare
Stick 4	1 L 5 S	1 L + 5 S	210.0 mm spare
Stick 5	4 S	0 L + 4 S	1479.6 mm spare

Finished-piece check:

Long: 8

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Short: 16

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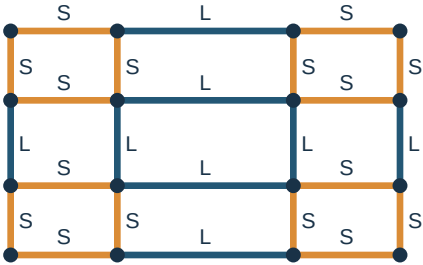
Assembly check: 30 mm insertion at each end. Short-section exposed span = 329.6 mm; long-section exposed span = 820.0 mm. Finished gate opening = 800 mm square.

For a hand saw, use the conservative stock-layout CSV linked in the guide; never add saw kerf to a finished part dimension.

800 mm opening / 18-inch paper

Outside paper square: 1714.4 mm. Buy 4 sticks of 10-ft nominal 1-inch PVC for the gate grid.

Mark	Cut quantity	Finished length (mm)	Decimal inches
L	8	880.0	34.6457
S	16	237.2	9.3386



Front grid / schematic, not to scale

Ratcheting cutter / stick-by-stick allocation

Blue = long (L). Orange = short (S). Gray = spare after 10 mm cleanup reserve. Numbers are quantities.

Stick 1	2 L	4 S	2 L + 4 S	329.2 mm spare
Stick 2	2 L	4 S	2 L + 4 S	329.2 mm spare
Stick 3	2 L	4 S	2 L + 4 S	329.2 mm spare
Stick 4	2 L	4 S	2 L + 4 S	329.2 mm spare

Finished-piece check:

Long: 8

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Short: 16

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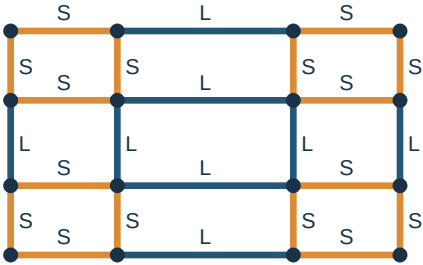
Assembly check: 30 mm insertion at each end. Short-section exposed span = 177.2 mm; long-section exposed span = 820.0 mm. Finished gate opening = 800 mm square.

For a hand saw, use the conservative stock-layout CSV linked in the guide; never add saw kerf to a finished part dimension.

800 mm opening / 12-inch paper

Outside paper square: 1409.6 mm. Buy 3 sticks of 10-ft nominal 1-inch PVC for the gate grid.

Mark	Cut quantity	Finished length (mm)	Decimal inches
L	8	880.0	34.6457
S	16	84.8	3.3386



Ratcheting cutter / stick-by-stick allocation

Blue = long (L). Orange = short (S). Gray = spare after 10 mm cleanup reserve. Numbers are quantities.

Stick 1	3 L	4 S	3 L + 4 S	58.8 mm spare
Stick 2	3 L	4 S	3 L + 4 S	58.8 mm spare
Stick 3	2 L	8 S	2 L + 8 S	599.6 mm spare

Finished-piece check:

Long: 8

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Short: 16

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Assembly check: 30 mm insertion at each end. Short-section exposed span = 24.8 mm; long-section exposed span = 820.0 mm. Finished gate opening = 800 mm square.

For a hand saw, use the conservative stock-layout CSV linked in the guide; never add saw kerf to a finished part dimension.